

THE HECKE MODULE OF THE COMPACT F_4 AT LEVEL 3: IDENTIFICATION OF ITS FIFTEEN EIGEN-SYSTEMS AND THREE NEW STABLE FORMS

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ABSTRACT. The compact F_4 over $\mathbb{Z}$ has, at the parahoric level Q_3 attached to the first fundamental coweight, a class set of 15 elements (Part I). We compute the two Hecke operators $T(\omega_1^\vee)(2)$ and $T(\omega_4^\vee)(2)$ on it—the second by nine explicit isomorphisms and a forced completion—and match each of the eight joint eigen-systems, at the prime 2, with an Arthur parameter built from forms on smaller groups: the lifts of Δ and of the newform in $S_{12}(\Gamma_0(3))$; the parameter 3.8.a.a[3] on Sp_6 , which carries the Eisenstein prime 41; 3.6.a.a $\oplus$ [4] on Sp_6 paired with Δ , which explains a congruence modulo 13 as the Eisenstein congruence of 3.6.a.a; two compact G_2 -forms of weight $2\omega_2$ and level 3, computed on G_2 and checked with both of its Hecke operators; and a compact $\mathrm{SO}(7)$ -form of weight $(3, 1, 1)$ and level 3 on the genus of E_7 , found by a computation in which all three of its Hecke eigenvalues at 2 agree with values predicted from the F_4 data, and whose Frobenius polynomial factors as 3.6.a.a times a stable genus-2 parameter of weight $\mathrm{Sym}^2 \otimes \det^6$ with $\lambda(2) = -12$ —the parameter predicted from F_4 before it was found. The two G_2 -forms and the genus-2 parameter admit no endoscopic description of the expected shapes. Every congruence in the module is explained either as a transported classical Eisenstein congruence or as level-raising at 3.

1. INTRODUCTION

In Part I [1] we computed the class set of the compact F_4 over $\mathbb{Z}$ at the parahoric level Q_3 (the stabiliser of a 6-dimensional null subspace of the Albert lattice modulo 3): fifteen classes, nine on the standard Albert lattice $J_{\mathbb{Z}}$ and six on the Elkies–Gross isotope J_E , and the Hecke operator $T(\omega_1^\vee)(2)$ on it, with integer spectrum

$$139230, \quad 8631^{(5)}, \quad 15057^{(2)}, \quad 3087^{(2)}, \quad 2331, \quad 567^{(2)}, \quad 207, \quad -585.$$

The Eisenstein primes 691, 73 and 41 predicted by the mass formula were confirmed there at the form level, and the eigenvalues 8631 and 15057 were identified as the theta lifts of Δ and of the unique newform 3.12.a.a in $S_{12}(\Gamma_0(3))$, which explains 691 and 73 as classical Eisenstein congruences transported to F_4 . The present paper identifies the remaining five eigenvalues.

The method is elementary and, we think, of independent use. Gross’s Satake dictionary expresses $T(\omega_1^\vee)(p)$ and $T(\omega_4^\vee)(p)$ as linear functions of the characters of the 26- and 52-dimensional representations of F_4 evaluated at the Satake parameter; for a parameter that factors through a subgroup $H \times H' \subset F_4$ these characters are polynomials in the Hecke eigenvalues of the pieces. Given a candidate shape, the two F_4 -eigenvalues therefore either determine the eigenvalues of the pieces or overdetermine them. We enumerate shapes, solve, and then *compute the predicted pieces on their own groups*: compact G_2 in weight $2\omega_2$, the non-split $\mathrm{SO}(5)$ on the genus of A_5 , and the split $\mathrm{SO}(7)$ on the genus of E_7 , with two or three Hecke operators each. A parameter is accepted only when every invariant available agrees. For -585 this produced a prediction—a genus-2 parameter with spinor eigenvalue -12 at 2, paired with 3.6.a.a—that was found, exactly, inside an $\mathrm{SO}(7)$ -form some sessions later.

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Remark 1.1 (Convention). Throughout this paper, to *identify* an eigen-system with a parameter means that the Satake parameter at 2, as seen by the two operators $T(\omega_1^\vee)(2)$ and $T(\omega_4^\vee)(2)$ (§4.1), agrees with that of the named parameter, and where a parameter is built from forms on a smaller group, that those forms have been computed there and their own Hecke eigenvalues at 2 agree. No Hecke operator of F_4 at a prime other than 2 has been computed, so every identification below is a match at one prime, with two invariants; we regard this as strong evidence and not as proof. At level one the identification of $\Theta(\Delta)$ has been checked at 5 on F_4 itself (§7); on the G_2 side the identification of the 9/4-form is verified at 5 (§8.2); the level- Q_3 predictions at 5 of §8.2 remain to be tested.

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2. THE CLASS SET AND THE FIRST OPERATOR

2.1. Level one (recollection). The compact F_4 over $\mathbb{Z}$ is $\text{Aut}(J_{\mathbb{Z}})$ for the integral Albert algebra $J_{\mathbb{Z}} = \text{Her}_3(\mathcal{O})$, and its level-one class set has two elements [10, 8]: $J_{\mathbb{Z}}$, with $|\text{Aut}| = 4180377600 = 6|W(E_8)|$, and the Elkies–Gross isotope J_E , with $|\text{Aut}| = 634023936 = 3|{}^3D_4(2)|$; the mass is $691/380414361600$. Their theta series are $E_{12} \pm (\text{const}/691)\Delta$, so the level-one Hecke module is $\mathbb{C}\mathbf{1} \oplus \mathbb{C}\Theta(\Delta)$ [26], and the congruence mod 691 is Ramanujan’s. We enumerated the 2-neighbours of both lattices for the two maximal parabolics (rank-one trace-zero points of $J/2J$, 69615 of them, in five Aut-orbits of sizes 48600, 14175, 6075, 405, 360; and 6-dimensional null subspaces for the Heisenberg parabolic) and classified each neighbour by explicit isomorphism. The Brandt matrices at $p = 2$ are

$$T(\omega_4^\vee)(2) = \begin{pmatrix} 1264320 & 7646400 \\ 1159704 & 7751016 \end{pmatrix}, \quad T(\omega_1^\vee)(2) = \begin{pmatrix} 25830 & 113400 \\ 17199 & 122031 \end{pmatrix},$$

with eigenvalues $\{8910720, 104616\}$ and $\{139230, 8631\}$. Both cuspidal eigenvalues agree with the Satake-parameter prediction from $\tau(2) = -24$ for the lift of Δ via $\iota : \text{Sp}_6 \times \text{SL}_2 \rightarrow F_4$ with the principal SL_2 (for $T(\omega_1^\vee)$: $N_2 - (1 + 2 + \dots + 2^5)(1 + 2^{11} - \tau(2)) = 139230 - 63 \cdot 1971 = 15057$ at level 3 below, and 8631 at level one). The 819 rank-one elements of J_E that define its relation to the Leech lattice [8] appear as the orbit of size 360; each has exactly 72 standard neighbours.

2.2. Level Q_3 . Let Q_3 be the parahoric at 3 attached to $\omega_1^\vee$ (stabiliser of a 6-dimensional null subspace of $J/3J$). The class set has 15 elements: nine on $J_{\mathbb{Z}}$ with stabiliser orders 288, 1296, 1944, 3024, 15552, 23328, 72576, 645120, 1244160, six on J_E with 36, 288, 648, 972, 2304, 9072; and

$$\sum_x \frac{1}{|\Gamma_x|} = \frac{2068163}{52254720} = \frac{691}{380414361600} \cdot 21789040,$$

21789040 being the number of Q_3 -cosets. The matrix of $T(\omega_1^\vee)(2)$ (Appendix A) has row sums 139230, is mass-weighted symmetric, and has integer spectrum

$$139230, \quad 8631^{(5)}, \quad 15057^{(2)}, \quad 3087^{(2)}, \quad 2331, \quad 567^{(2)}, \quad 207, \quad -585.$$

Here $E = 41 \cdot 73 \cdot 691$, as the mass formula [1, (1)] predicts $(41 \mid 3^8 - 1, 73 \mid 3^{12} - 1, 691 \text{ from } \zeta(-11), \text{ with 5 and 7 cancelling})$, and the form-level test passes at exactly these primes: 691 on the Δ -lift eigenspace, 73 on the 15057-eigenspace, 41 on the 2331-eigenspace, and at no other prime (the eigenvalue coincidences at 2161, 53, 31, 71, 19, 271, 13, 239 all fail the test).

3. THE SATAKE DICTIONARY AND THE CLASSICAL LIFTS

3.1. Identification of the new eigenvalues. The theta correspondence for $F_4 \times \mathrm{PGL}_2$ [26] lifts a newform f of weight 12 and level 3 to eigenvalue $N_2 - 63(1 + 2^{11} - a_2(f))$. The space $S_{12}(\Gamma_0(3))$ has a unique newform, 3.12.a.a, with $a_2 = 78$, $a_3 = -3^5$; it lifts to 15057, and $1 + p^{11} - a_p(f) \equiv 0 \pmod{73}$ for $p = 2, 5, 7$: the prime 73 on F_4 is the classical Eisenstein congruence of 3.12.a.a, transported, exactly as 691 is Ramanujan's. The remaining eigenvalues are *not* such lifts: 3087, 2331, 567 would need $a_2 = -112, -124, -152$, outside the Ramanujan bound $2 \cdot 2^{11/2}$, and 207, -585 are not of the required shape at all.

The dictionary behind these formulas is $T(\omega_1^\vee)(p) = p^8 \chi_{26}(s) - (1 + p^4)$, where s is the Satake parameter and χ_{26} the character of the 26-dimensional representation; for a lift via $\mathrm{Sp}_6 \times \mathrm{SL}_2$ one has $26 = (14, 1) \oplus (6, 2)$ with e_p principal in Sp_6 , which reproduces 139230, 8631 and 15057 from $a_2 = 2049, -24, 78$. The same dictionary through the subgroup $G_2 \times \mathrm{SL}_2 \subset F_4$, under which $26 = (7, 3) \oplus (1, 5)$, with the SL_2 principal, sends a G_2 eigenform with $T(\omega_1^\vee)(2) = t$ (so $\mathrm{Tr} V_7 = (t + 1)/8$) to

$$p^8 \left(\frac{t+1}{8} (p^{-1} + 1 + p) + (p^{-2} + p^{-1} + 1 + p + p^2) \right) - (1 + p^4) = 112(t + 1) + 1967 \quad (p = 2).$$

The compact G_2 at level 3 has trivial-weight cuspidal eigenforms with $t = 9$ [15], and $112 \cdot 10 + 1967 = 3087$; this is a coincidence: a trivial-weight G_2 form has Hodge weights $\{0, \pm 1, \pm 2, \pm 3\}$ on the 7, so $(7, 3) \oplus (1, 5)$ does not have the infinitesimal character of the trivial representation of F_4 , and the correct lifts through G_2 involve the weight $2\omega_2$ (§8.1). (The trivial-weight form is itself the G_2 -lift of 3.6.a.a: $\chi_7 = t_{3.6.a.a} \chi_{[2]} + \chi_{[3]} = 5/4$.) Two facts about the compact G_2 at 3 are used below. First, at all three parahorics at 3—the two maximal ones (stabilisers of a null line, resp. a null plane, modulo 3) have class number 2, stabiliser orders $\{432, 36\}$ and $\{48, 108\}$, mass $13/432$ and a single cuspidal eigen-system $(T_2, T_5) = (9, 810)$; the Iwahori has class number 3, mass $13/108$, and the same eigen-system with multiplicity 2—no trivial-weight form with $T_2 = -12$ exists, so 207 is not a $G_2 \times \mathrm{SL}_2$ lift with trivial SL_2 -parameter (which would need $96(t + 1) + 1263 = 207$, i.e. $t = -12$). Second, the dictionary predicts $T(\omega_1^\vee)(5) = 5^8 \left(\frac{811}{125} \cdot \frac{31}{5} + \frac{781}{25} \right) - 626$ on a $G_2 \times \mathrm{SL}_2$ lift of the $t = 9$ form; the corresponding predictions for the parameters identified below are the test of Remark 1.1.

3.2. Temperedness. With $\chi_{26}(s) = (\lambda + 17)/256$, a unitary Satake parameter forces $|\chi_{26}(s)| \leq 26$, i.e. $\lambda \in [-6673, 6639]$. The two identified PGL_2 -lifts violate this ($\chi_{26} = 33.8$ and 58.9), as non-tempered Arthur parameters must; the G_2 -lift 3087 ($\chi_{26} = 12.1$) and the then-unidentified 2331, 567, 207, -585 (9.2, 2.3, 0.9, -2.2) all satisfy it. The 41-carrier is therefore consistent with being tempered, and if it is a lift from a smaller group the lifting parameter cannot involve a principal SL_2 of the type that produced 8631 and 15057.

4. THE SECOND HECKE OPERATOR

4.1. A second Hecke operator. We computed $T(\omega_4^\vee)(2)$ on the 15 classes. Its 8,910,720 neighbours per class are indexed by the 69615 rank-one points of $J/2J$ and 128 lifts of each; for each of the five $\mathrm{Aut}(J_{\mathbb{Z}})$ -orbits of points, the stabiliser of a representative point permutes its lifts, and a count on the level-3 orbits is constant on a stabiliser-orbit of lifts, so one explicit isomorphism per stabiliser-orbit suffices: the 128 lifts fall into 2, 2, 1, 2, 2 orbits, i.e. *nine* isomorphisms (not 640), found and verified exactly as for $T(\omega_1^\vee)(2)$. This gives the nine standard rows; the (E, Z) block follows by weighted symmetry, and the (E, E) block is then *forced*: the conditions $[T(\omega_1^\vee)(2), T(\omega_4^\vee)(2)] = 0$, weighted symmetry and the row sums have a unique solution. Checks: all 225 entries are non-negative integers; every row sums to 8,910,720; every standard row splits 1,264,320 standard + 7,646,400 exotic and every exotic row 1,159,704 + 7,751,016, exactly the

level-one Brandt matrix, although the exotic rows were not computed from neighbours. The joint eigen-systems are

$T(\omega_1^\vee)(2)$	$T(\omega_4^\vee)(2)$	mult.	identification
139230	8910720	1	constant
15057	336870	2	3.12.a.a[6] $\oplus$ [9] $\oplus$ [5] (predicted 336870)
8631	104616	5	$\Delta[6] \oplus [9] \oplus [5]$ (predicted 104616)
3087	23040	2	$G_2 \times \mathrm{SO}(3)$ -lift, G_2 weight $2\omega_2$, $T_2 = 9$ (§8.1)
2331	11916	1	3.8.a.a[3] via $\mathrm{Sp}_6 \times \mathrm{SL}_2$ (predicted 11916)
567	-14040	2	$G_2 \times \mathrm{SO}(3)$ -lift, G_2 weight $2\omega_2$, $T_2 = -27/2$ (§8.1)
207	28800	1	3.6.a.a $\oplus$ [4] on Sp_6 , Δ on SL_2 (§6.3)
-585	9000	1	$\wedge_0^2 F \oplus F[2]$, F an $\mathrm{SO}(7)$ -form of weight $(3, 1, 1)$ (§9.2)

With $T(\omega_4^\vee)(p) = p^{11}\chi_{52}(s_p) - p^8\chi_{26}(s_p) - p^6(p^4 + 1)$ and the restriction of the adjoint representation to $\mathrm{Sp}_6 \times \mathrm{SL}_2$, $52 = (21, 1) \oplus (1, 3) \oplus (14', 2)$, the three identified PGL_2 -lifts were predicted before the computation and all three predictions hold exactly; in particular the identification of the 41-carrier with 3.8.a.a[3] is now confirmed by two independent operators. Through $G_2 \times \mathrm{SL}_2$, $52 = (14, 1) \oplus (1, 3) \oplus (7, 5)$, the eigenvalue 23040 determines the adjoint character of the level-3 form of G_2 at 2: $\chi_{14}(s_2) = 7/64$, consistent with a tempered Satake parameter. All twelve congruences of the tables above hold for $T(\omega_4^\vee)(2)$ as well, including those modulo 5 and 7, which are therefore genuine.

Genus-2 shapes and the prediction for -585. For the genus-2 shapes $\psi = \wedge_0^2 F \oplus F \otimes g \oplus [1] \oplus F[2] \oplus g[2]$, the pair (χ_{26}, χ_{52}) determines the normalised traces $(\mathrm{tr} F(s_2), \mathrm{tr} \wedge_0^2 F(s_2))$ up to a quadratic ambiguity. For 567 and 207 no shape admits a unitary solution: *neither is a genus-2 lift of this kind.* For -585 the shape with $g = 3.6.a.a$ gives the unitary, exactly rational solution $\mathrm{tr} F(s_2) = -3 \cdot 2^{-7/2}$, $\mathrm{tr} \wedge_0^2 F(s_2) = -11/16$, i.e. an *integral* spinor eigenvalue $\lambda_F(2) = -6$ if F has weight $\mathrm{Sym}^6 \otimes \det^3$ (resp. -12 for $\mathrm{Sym}^2 \otimes \det^6$); the other shapes give irrational or non-unitary values. We therefore predict: *there is a genus-2 Siegel cusp form of level dividing 3 and weight $\mathrm{Sym}^6 \otimes \det^3$ (or $\mathrm{Sym}^2 \otimes \det^6$) with spinor Hecke eigenvalue -6 (resp. -12) at 2, and the -585-form is its lift with 3.6.a.a.* We tested the level. Through $\mathrm{SO}(5) \sim \mathrm{PGSp}_4$, forms of paramodular level 3 correspond to algebraic modular forms on the genus of the root lattice A_5 (even, quinary, half-discriminant 3), which has a single class, with $\mathrm{SO}(A_5) \cong S_6$; the Siegel weights $\mathrm{Sym}^j \otimes \det^k$ correspond to the $\mathrm{SO}(5)$ -representations of highest weight $((j + 2k - 6)/2, j/2)$, e.g. $(3, 3)$ for $\mathrm{Sym}^6 \otimes \det^3$. By the Weyl character formula the invariant spaces are 0 for every candidate weight: *there is no such form of paramodular level 3.* At the smaller level K' given by the stabiliser of an isotropic line in $A_5/3A_5$ (thirty non-radical isotropic lines in two orbits, with stabilisers of orders 36 and 72) the spaces are non-zero, and we computed the 2-neighbour operator there. We realised $V_{(3,3)}$ as the 84-dimensional Casimir eigenspace in $\mathrm{Sym}^3(\wedge^2 V)$, found the fifteen 2-neighbours of A_5 with explicit rational isometries onto A_5 , and transported the level structure (checks: the representation is multiplicative on rational isometries, its image preserves the invariant subspaces, and on trivial weight T_2 has eigenvalues 15 and 6). In weight $V_{(3,3)}$ the space has dimension $3 + 1 = 4$ and

$$T_2 \text{ has eigenvalues } 6, \quad -\frac{3}{4}, \quad \frac{3}{2}, \quad \frac{3}{2}.$$

Since the p -neighbour operator corresponds to the (minuscule) standard representation of the dual group Sp_4 , a tempered eigenform has $T_2 = 2^{3/2} \mathrm{tr} F(s_2)$, and the predicted $\mathrm{tr} F(s_2) = -3 \cdot 2^{-7/2}$ gives $T_2 = -3/4$, which occurs. The prediction also fixes $\mathrm{tr} \wedge_0^2 F(s_2) = -11/16$, and we tested it with the operator T_P attached to totally isotropic planes of $A_5/2A_5$: there are 30 such neighbours (all isometric to A_5 ; the enumeration also returns the 15 ordinary neighbours, separated by their elementary divisors), T_P commutes with T_2 , and since its Satake transform

lies in the span of $\chi_{\wedge_0^2}$ and 1 with leading coefficient $2^{(\rho, \lambda)} = 4$, the constant function calibrates it: $T_P = 4\chi_{\wedge_0^2}(s_2) - 1$. On $V_{(3,3)}$ the joint eigen-systems are $(T_2, T_P) = (6, \frac{51}{8}), (-\frac{3}{4}, \frac{21}{16}), (\frac{3}{2}, -\frac{57}{8})^{(2)}$. *The second test fails:* the form with $T_2 = -3/4$ has $\text{tr } \wedge_0^2 F(s_2) = 37/64$, not $-11/16$, so its match in the first invariant was a coincidence. Conversely, lifting each of the three genus-2 eigen-systems through the shape with $g = 3.6.a.a$ gives $(T(\omega_1^\vee)(2), T(\omega_4^\vee)(2)) = (711, 5112), (-261, 252)$ and $(-585, 16776)$: the last reproduces the first F_4 eigenvalue exactly but not the second (9000), because the first F_4 operator constrains the Satake parameter of F only along a line. None of the genus-2 forms of weight $\text{Sym}^6 \otimes \det^3$ at level K' is the one predicted; the predicted parameter is found in §§9.2–10, in its $\text{Sym}^2 \otimes \det^6$ version, inside an endoscopic form on $\text{SO}(7)$.

5. CONGRUENCES IN THE MODULE

5.1. Congruences between cuspidal eigenforms. Two eigenspaces of $\mathbb{T}$ are congruent modulo ℓ when their saturated integer eigenlattices meet modulo ℓ ; this is a congruence of eigenvectors, hence of eigen-systems for the whole Hecke algebra. Applied to the 23 rational eigenforms of the Niemeier genus, the test recovers Harder’s congruence modulo 41 between the eigenforms with $\lambda_2 = 21600$ and -7920 [6], together with the propagation of the Bernoulli primes among the lifts and a dozen congruences modulo 11, 13, 17, 19, 23 between pairs of eigenforms. Applied to the F_4 module at level Q_3 (omitting $\ell = 2, 3$) it gives the three Eisenstein congruences of the theorem and the following cuspidal ones:

eigenvalues	ℓ	remark
15057, 8631	17 (two-dimensional)	the classical congruence $\Delta \equiv 3.12.a.a$, $\tau(2) - a_2 = -102 = -6 \cdot 17$, transported
8631, 3087	11 (two-dimensional)	the 3087-forms are $G_2 \times \text{SO}(3)$ -lifts of weight- $2\omega_2$ G_2 -forms (§8.1)
8631, 207	13	cyclotomic level-raising at 3 ($13 \mid 3^3 - 1$); the 207-form is not built from PGL_2
207, -585	11	
15057, 3087; 15057, 567; 8631, 2331; 3087, 567	5	confirmed by $T(\omega_4^\vee)(2)$ (§4.1)
2331, 567	7	confirmed by $T(\omega_4^\vee)(2)$

The congruences modulo 11 and 13 between the lift of Δ and eigenforms that are not lifts are the F_4 analogues of Harder’s congruence: if the 3087- and 207-forms are lifts from Sp_6 or $\text{Spin}(9)$, these primes should divide the corresponding L -values.

5.2. Level-raising at 3. Every congruence between the Δ -lift, which is old (it has a vector fixed by the hyperspecial subgroup), and a form new at 3 should be a level-raising congruence: a prime ℓ at which two eigenvalues e_i, e_j of the Frobenius at 3 of $\psi_\Delta = \Delta[6] \oplus [9] \oplus [5]$ on the 26 satisfy $e_i \equiv 3e_j \pmod{\ell}$ without $e_i = 3e_j$. Computing all such pairs in $\mathbb{Q}(\alpha_3(\Delta))$ gives the level-raising primes 5, 7, 11, 13, 17, 19, 41, 83, 137, 181, 691, $\dots$ of ψ_Δ at 3, and every cuspidal congruence of the Δ -lift is of this kind:

- mod 17 (with the 3.12.a.a-lift): internal to the Δ -block, Ribet’s condition $\tau(3)^2 \equiv (1 + 3)^2 3^{10}$;
- mod 11 (with the G_2 -lift): between the Δ -block and a trivial block, $\tau(3) \equiv 3^3 + 3^8 \pmod{11}$;
- mod 13 (with the 207-form) and mod 5 (with the 2331-form): *cyclotomic*, e_i/e_j a power of 3 with $13 \mid 3^3 - 1$, resp. $5 \mid 3^4 - 1$, independent of $\tau(3)$; these are primes of the parahoric index $[K : Q_3] = 21789040 = 2^4 \cdot 5 \cdot 7 \cdot 13 \cdot 41 \cdot 73$.

For the constant function the same cyclotomic condition holds at 5, 7, 13, and the congruences do *not* occur, because the level-one mass has 5, 7, 13 in its denominator and they cancel in E ([1, Thm. 2.1]); for the Δ -lift the level-one congruence module contains only 691, nothing absorbs 13 or 5, and the congruences occur. We therefore propose the following organising principle, which accounts for every congruence in the F_4 level- Q_3 module that involves a known form: *a congruence is either Eisenstein, at a prime of E , and then a transported classical Eisenstein congruence; or level-raising of an old form at 3, at a prime of the parahoric index not absorbed at level one or at a prime where the old form's Satake parameter at 3 satisfies a level-raising condition.*

6. SEARCHES THROUGH PARAMETERS OF $\mathrm{Sp}_6 \times \mathrm{SL}_2$ AND $\mathrm{Spin}(9)$

6.1. An Arthur-parameter search. A parameter ψ for F_4 that is a sum of blocks $\pi[d]$ (π cuspidal on PGL_n , $[d]$ the d -dimensional representation of the Arthur SL_2) contributes to the trivial-weight spectrum only if its infinitesimal character at ∞ is that of the trivial representation, which on the 26-dimensional representation is the multiset $\{-8, \dots, 8\} \cup \{-4, \dots, 4\}$ (the principal SL_2 acts on 26 as $[17] \oplus [9]$). We enumerated every ψ built from the trivial representation, the newforms of level 1 and 3 and weight ≤ 16 (computed from q -expansions: of level 3 they are 3.6.a.a, 3.8.a.a, two of weight 10, 3.12.a.a, three of weight 14, two of weight 16), and their symmetric squares, satisfying this constraint, and evaluated $T(\omega_1^\vee)(2) = 256\chi_{26}(s_2) - 17$ on each. There are 117; all eigenvalues are $\equiv 0 \pmod{9}$, and 62 of the 1479 residues-0 slots in the tempered range are occupied, so a random integer has about a 4% chance of being hit. The search reproduces $139230 = [17] \oplus [9]$, $8631 (\Delta[6] \oplus [9] \oplus [5])$ and $15057 (3.12.a.a[6] \oplus [9] \oplus [5])$, and finds

$$2331 : \quad \psi = f[2] \oplus f[4] \oplus \mathrm{Sym}^2 f[3] \oplus [5], \quad f = 3.8.a.a \ (a_2 = 6).$$

This ψ is the image under $\mathrm{Sp}_6 \times \mathrm{SL}_2 \subset F_4$ of the Sp_6 -parameter $f[3]$ with the Arthur SL_2 embedded diagonally: $26 = (14, 1) \oplus (6, 2)$ restricts to $\wedge_0^2(f[3]) \oplus f[3] \otimes [2] = ([5] \oplus \mathrm{Sym}^2 f[3]) \oplus (f[4] \oplus f[2])$. It is confirmed by the prime it carries: $1 + 2^7 - a_2(f) = 123 = 3 \cdot 41$, $41 \mid 3^8 - 1$, and $f \equiv 1 \oplus \chi^7 \pmod{41}$ is the classical Eisenstein congruence of the weight-8 newform of level 3; substituting $1 \oplus \chi^7$ for f in ψ gives exactly the multiset of the principal parameter, so $\psi \equiv [17] \oplus [9] \pmod{41}$. *The prime 41 on the 2331-form is the Eisenstein congruence of 3.8.a.a, transported to F_4 , as 691 on 8631 is Ramanujan's and 73 on 15057 is that of 3.12.a.a.* This also explains the theta-invisibility: the form is attached to weight 8, not 12.

The only parameter found for 567 involves four distinct newforms $(3.6.a.a[4] \oplus 3.8.a.a[2] \oplus \Delta[2] \oplus 3.16[2] \oplus [5] \oplus [1])$; its image would contain four commuting SL_2 's and a nontrivial commuting Arthur SL_2 , of total rank $5 > 4 = \mathrm{rk} F_4$, so it is not realisable and the hit is a coincidence. No parameter of this kind gives 207 or -585 . Hence $567^{(2)}$, 207, -585 do not come from forms on PGL_2 of level dividing 3: their parameters involve a stable form on G_2 , a non-lifted Siegel form of genus 2, or are stable on F_4 .

The cuspidal congruences are consistent with this. The congruence $\Delta \equiv 3.12.a.a \pmod{17}$ is Ribet level-raising at 3: $\tau(3)^2 - (1 + 3)^2 3^{10} = -2^7 \cdot 3^4 \cdot 5 \cdot 17$.

6.2. Exhaustive genus-2 and triality sweeps. On the genus of A_5 the two levels at 3 are the only ones (the quadratic space $A_5/3A_5$ has a one-dimensional radical and an anisotropic quotient of type $O^-(4, 3)$: its 30 isotropic lines lie over 10 points), so we computed *every* admissible genus-2 weight at both levels with both operators T_2, T_P and lifted each eigen-system to F_4 through its shape. Paramodular level: all spaces are 0. Level K' : weights $(2, 2), (3, 2), (3, 3), (4, 1)$ of dimensions 3, 2, 4, 3 (weight $(1, 1)$ is 0). Two of the nine eigen-systems are Yoshida lifts, $(T_2, T_P) = (-\frac{3}{2}, \frac{3}{4}) = Y(3.8.a.a, 3.6.a.a)$ in weight $\mathrm{Sym}^4 \otimes \det^3$ and $(-\frac{15}{2}, \frac{39}{4}) = Y(3.10.b, 3.6.a.a)$

in weight $\text{Sym}^4 \otimes \det^4$; as a check, their F_4 -lifts coincide, $(-585, 3816)$, because they are built from the same three newforms. The other seven are not Yoshida or Saito–Kurokawa lifts of forms of level dividing 3 and are presumably stable. *No lift matches any of* $(567, -14040)$, $(207, 28800)$, $(-585, 9000)$.

We also searched the parameters through $\text{Spin}(8) \supset \text{SL}_2^4$, where $26 = 1 + 1 + 8_v + 8_s + 8_c$ and each 8 is a sum of tensor products of pairs of the four factors: filling the factors with our newforms and the Arthur SL_2 gives exactly two admissible parameters up to triality; the one built from 3.6.a.a, 3.8.a.a, 3.10.b and [2] again gives $(-585, 3816)$, and the other nothing. The first operator thus returns -585 for several unrelated parameters—it sees only χ_{26} , which is linear in the parameter’s traces—while the second separates them, and none has $T(\omega_4^\vee)(2) = 9000$. At this stage none of 567, 207, -585 was a lift from any family testable with forms of level dividing 3 on the genera of A_5 and through $\text{Spin}(8)$; they are resolved in §§6.3, 8.1 and 9.2.

6.3. A guess-and-check engine. Any parameter through $\text{Sp}_6 \times \text{SL}_2$ is determined by an Sp_6 -parameter P and an SL_2 -parameter Q , and then $26 = \wedge_0^2 P \oplus P \otimes Q$ and $52 = \text{Sym}^2 P \oplus \text{Sym}^2 Q \oplus \wedge_0^3 P \otimes Q$; through $\text{Spin}(9) \supset \text{Spin}(5) \times \text{Spin}(4)$ one has $26 = 1 \oplus \wedge_0^2 F \oplus g \otimes h \oplus F \otimes g \oplus F \otimes h$ and $52 = \text{Sym}^2 F \oplus \wedge_0^2 F \otimes g \otimes h \oplus \text{Sym}^2 g \oplus \text{Sym}^2 h \oplus F \otimes g \oplus F \otimes h$. Both F_4 -eigenvalues follow from the Satake eigenvalues of the pieces. We generated every P of dimension 6 built from newforms of level 1 and 3 (we also included those of level 9—twists by the character of conductor 3, CM forms by $\mathbb{Q}(\sqrt{-3})$, four quadratic pairs—although, the forms being fixed by a parahoric and hence Iwahori-spherical at 3, they cannot occur; none enters any identification), Arthur blocks $[d]$, $f[d]$, $\text{Sym}^2 f[2]$, $\text{Sym}^3 f$, $f \otimes \text{Sym}^2 h$ and the genus-2 eigen-systems of §4.1, every Q , and also triple products through $\text{Spin}(9) \supset \text{SL}_2^3$, with the rank condition imposed. There are 351 parameters with the infinitesimal character of the trivial representation. The engine reproduces 8631, 15057 and 2331 exactly (the G_2 -lift 3087 lies outside it) and finds, uniquely,

$$207 : \quad \psi = 3.6.a.a[4] \oplus [5] \oplus [1] \oplus (3.6.a.a \otimes \Delta) \oplus \Delta[4],$$

the image of the Sp_6 -parameter $3.6.a.a \oplus [4]$ with Δ on the SL_2 factor; both $T(\omega_1^\vee)(2) = 207$ and $T(\omega_4^\vee)(2) = 28800$ are reproduced. The earlier searches missed it because it contains the four-dimensional tensor block $3.6.a.a \otimes \Delta$. Its congruence with the Δ -lift modulo 13 is now explained classically: $1 + 2^5 - a_2(3.6.a.a) = 39 = 3 \cdot 13$ with $13 \mid 3^6 - 1$, so $3.6.a.a \equiv 1 \oplus \chi^5 \pmod{13}$, and substituting this in ψ turns $3.6.a.a[4] \oplus [5] \oplus [1]$ into $[9] \oplus [5]$ and $(3.6.a.a \otimes \Delta) \oplus \Delta[4]$ into $\Delta[6]$, the parameter of the Δ -lift: $207 \equiv 8631 \pmod{13}$ is the Eisenstein congruence of 3.6.a.a, transported, in the same way as 691, 73 and 41. For 567 and -585 no candidate reproduces both eigenvalues; twelve reproduce $T(\omega_1^\vee)(2) = -585$ alone, all containing 3.6.a.a, which is again the weakness of the first operator.

6.4. Exclusions for 567 and 207. Before their identification (§§6.3, 8.1) we tested $567^{(2)}$ and 207 against two further families of lifts; the exclusions remain of interest as uniqueness statements. (i) *Stable Sp_6 -parameters F* (6-dimensional, Hodge weights $\pm a/2, \pm b/2, \pm c/2$) through $\text{Sp}_6 \times \text{SL}_2$: the shapes compatible with the infinitesimal character are $\wedge_0^2 F \oplus F[2]$ with $(a, b, c) \in \{(11, 5, 3), (9, 7, 5)\}$ and $\wedge_0^2 F \oplus F \otimes h$ with h one of 3.6.a.a, 3.8.a.a, the two weight-10 forms, Δ , 3.12.a.a. Here (χ_{26}, χ_{52}) leaves one free parameter; requiring the eigenvalues of F on $\wedge^1, \wedge^2, \wedge^3$ to be rational integers after the arithmetic normalisation and the Satake parameter to be unitary leaves *no* candidate for 567 or 207 (and a one-parameter family for -585 , so this test does not decide -585). (ii) $\text{SL}_3 \times \text{SL}_3$: the only shapes compatible with the infinitesimal character are $\text{ad } \pi \oplus \pi[3] \oplus \pi^\vee[3]$ and $\text{ad}[3] \oplus [3] \otimes \pi \oplus [3] \otimes \pi^\vee$ with π of Hodge weights $\{3, 2, -5\}$, $\{5, -2, -3\}$, $\{4, 3, -7\}$ or $\{7, -3, -4\}$; none of these is symmetric about its centre, so by Clozel’s purity lemma no cuspidal π on PGL_3 has them, and the family is empty. Combined with §§6.1–4.1: $567^{(2)}$ and 207 are not lifts from PGL_2 (level dividing 3, weight ≤ 16),

$G_2 \times \mathrm{SL}_2$, genus-2 Siegel forms, stable Sp_6 -forms with rational eigenvalues, or PGL_3 . The parameter shapes not excluded are those through $\mathrm{Spin}(9)$ with a stable parameter of the group whose dual is $\mathrm{Spin}(9)$, lifts with non-rational Hecke fields, and stable parameters of F_4 . (Both were subsequently identified: 207 in §6.3 and 567 in §8.1, through parameter shapes outside the families listed here.)

6.5. Genus-2 candidates. If one of $567^{(2)}$, 207, -585 is a lift involving a genus-2 Siegel form F (spinor Hodge weights $\pm a/2, \pm b/2$), the only shapes compatible with the infinitesimal character of F_4 through $\mathrm{Sp}_6 \times \mathrm{SL}_2$ or $\mathrm{Spin}(9)$ are $\psi = \wedge_0^2 F \oplus F \otimes g \oplus [1] \oplus F[2] \oplus g[2]$ with

$$(F, g) \in \{(\mathrm{Sym}^2 \otimes \det^3, \Delta), (\mathrm{Sym}^2 \otimes \det^3, 3.12.a.a), (\mathrm{Sym}^4 \otimes \det^3, \text{wt } 10), \\ (\mathrm{Sym}^4 \otimes \det^4, 3.8.a.a), (\mathrm{Sym}^6 \otimes \det^3, 3.6.a.a), (\mathrm{Sym}^2 \otimes \det^6, 3.6.a.a)\},$$

F of level dividing 3 (the shape $1 \oplus \wedge_0^2 F \oplus [3] \oplus [1] \oplus 2F[2]$ has no solution). Of these six shapes exactly one occurs: the last, realised by the parameter G of §10 paired with 3.6.a.a; the others are excluded by the two-operator test of §4.1.

7. A SECOND PRIME ON F_4 : THE OPERATOR $T(\omega_1^\vee)(5)$ AT LEVEL ONE

The identifications of this paper are matches at the prime 2 (Remark 1.1). At level one the theta-lift identification of the cusp form with $\Delta[6] \oplus [9] \oplus [5]$ predicts, from $\tau(5) = 4830$ and the dictionary of §3.1, the eigenvalue

$$T(\omega_1^\vee)(5)|_{\Theta(\Delta)} = 336,244,104, \quad T(\omega_1^\vee)(5)|_1 = N_5 = 191,040,038,280,$$

the second being 5 times the number 38,208,007,656 of 6-dimensional sharp-null subspaces $S \subset J/5J$ (each has exactly five lifts, all Albert lattices). Direct enumeration is out of reach, so we computed the $\mathrm{Aut}(J_\mathbb{Z})$ -orbits of the subspaces S . For each sampled S the intersection lattice $M = J \cap J'$ (index 5^6 in J) has a small orthogonal group (orders 8–128 in general, against $|O(J_\mathbb{Z})| \approx 9.7 \cdot 10^{28}$), and $\mathrm{Stab}(S)$ is the set of its elements that are integral Jordan automorphisms of $J_\mathbb{Z}$, verified exactly (the sharp identity on all basis vectors and pairwise sums). Orbits are separated by isometry invariants of M and, where these coincide, by an exact isomorphism test (`qfisom` composed with $O(M)$ and the exact Jordan test). The mass formula is the stopping rule: $\sum_{\text{orbits}} |\mathrm{Stab}(S)|^{-1} = 38208007656/|\mathrm{Aut}(J_\mathbb{Z})| = 75809539/8294400$, and the entry $a = \#\{J_\mathbb{Z}\text{-type neighbours of } J_\mathbb{Z}\}$ is $|\mathrm{Aut}(J_\mathbb{Z})| \sum_{\text{orbits}} n_Z(S)/|\mathrm{Stab}(S)|$, with $n_Z(S)$ the number of lifts of S isomorphic to $J_\mathbb{Z}$.

From 10,238 uniformly sampled subspaces: 37 invariant classes, four of which are each exactly two orbits with isometric intersection lattices (separated only by the Jordan structure), with stabiliser orders from 2 to 192; together they account for mass $5239/576$, leaving an exact deficit $367939/8294400$, and give $\sum n_Z/|\mathrm{Stab}| = 3379/576$. Since $0 \leq n_Z \leq 5$, the eigenvalue $\lambda = a(1+r) - rN_5$ ($r = |\mathrm{Aut}(J_E)|/|\mathrm{Aut}(J_\mathbb{Z})|$) lies in

$$[-731,588,462, 336,244,104],$$

whose upper end is the prediction, attained exactly when every orbit not yet analysed has all five lifts of type $J_\mathbb{Z}$. The unanalysed subspaces are the 55 sampled ones whose intersection lattices have orthogonal groups of order $> 2 \cdot 10^5$ (≈ 50 expected from the deficit), and every one of them has $n_Z = 5$. As an independent check, the plain proportion of $J_\mathbb{Z}$ -type lifts among the sampled subspaces is 0.1346 ± 0.0016 , against 0.13322 predicted. We conclude $T(\omega_1^\vee)(5)$ has eigenvalue 336,244,104 on $\Theta(\Delta)$, as predicted. The one step not yet carried out by machine is the computation of the stabilisers of the large-symmetry orbits, whose masses must sum to $367939/8294400$; it requires an automorphism search on Albert lattices restricted to those preserving M , and would make the result a complete computation.

8. THE G_2 -LIFTS OF WEIGHT $2\omega_2$

8.1. The G_2 -lifts of weight $2\omega_2$. Through $G_2 \times \mathrm{SO}(3) \subset F_4$, with $26 = (7, 3) \oplus (1, 5)$ and $52 = (14, 1) \oplus (1, 3) \oplus (7, 5)$, and the Arthur SL_2 on the $\mathrm{SO}(3)$ factor, both F_4 -eigenvalues are *linear* in the characters (χ_7, χ_{14}) of a G_2 -parameter, so each open eigen-system determines them exactly. The infinitesimal character forces the G_2 Hodge weights $\{0, \pm 3, \pm 4, \pm 7\}$, which by the Weyl dimension formula ($\dim = uv(u+v)(v-u)(2u+v)(u+2v)/120$ for Hodge type (u, v)) is the weight $2\omega_2$, of dimension 77—not the trivial weight, and not $2\omega_1$ (27, type $(1, 4)$). One finds: $3087 \leftrightarrow (\chi_7, \chi_{14}) = (5/4, 7/64)$, $567 \leftrightarrow (-25/16, 329/128)$, both tempered, i.e. G_2 -eigenvalues $T(\omega_1^\vee)(2) = 8\chi_7 - 1 = 9$ and $-27/2$. We computed compact G_2 forms of weight $2\omega_2$ at all three levels at 3 (the representation realised as the top Casimir component of Sym^2 of the adjoint, built from the derivations of the octonions; the 126 neighbour maps are octonion automorphisms; trivial weight reproduces $\{126, 9\}$ and $\{126, 9, 9\}$; the operator preserves the invariant subspaces to 10^{-12}). At the line-parahoric level the space has dimension 5 and

$$T_2 \text{ has eigenvalues } -\frac{27}{2}, -\frac{27}{2}, \frac{9}{4}, 9, 9,$$

in exact correspondence, with multiplicities, with the five-dimensional part $3087^{(2)} \oplus 2331 \oplus 567^{(2)}$ of the F_4 module. The eigenvalue $9/4$ is the G_2 -lift of 3.8.a.a ($\chi_7 = \chi_{[2]}t + t^2 - 1 = 13/32$, $\chi_{14} = 0.8477$), whose $G_2 \times \mathrm{SO}(3)$ -lift reproduces *both* F_4 -eigenvalues $(2331, 11916)$, independently of its identification through Sp_6 in §6.1. We therefore identify $3087^{(2)}$ and $567^{(2)}$ as the $G_2 \times \mathrm{SO}(3)$ -lifts of the weight- $2\omega_2$ G_2 -forms of level 3 with $T_2 = 9$ and $-27/2$. The second invariants are checked with the second Hecke operator of G_2 . Among the 10207 lattices reached from the Coxeter order in two $\omega_1^\vee$ -steps, classified by elementary divisors, those of relative position $(\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 1, 1, 2, 2, 4)$ number $2016 = 2^5 \cdot 63$ and form the $\omega_2^\vee$ -double coset (the others: 1, 126, and 8064 of type $2\omega_1^\vee$). Its Satake transform lies in the span of $\chi_{14}, \chi_7, 1$ with leading coefficient $2^{\langle \rho, \omega_2^\vee \rangle} = 32$, and the constant function and the trivial-weight cusp form of level 3 calibrate it: $T(\omega_2^\vee)(2) = 32(\chi_{14} - \chi_7/4 - 1/2)$. On the five-dimensional weight- $2\omega_2$ space it commutes with T_2 and has eigenvalues $63/8, -45/2^{(2)}, 315/4^{(2)}$ on the forms with $T_2 = 9/4, 9, -27/2$, i.e. $\chi_{14} = 217/256, 7/64, 329/128$. The first is the value predicted for the G_2 -lift of 3.8.a.a, and the other two are exactly the values required by the F_4 eigen-systems of 3087 and 567. *Both F_4 eigenvalues of 3087 and 567 are thus reproduced by forms computed independently on G_2 , with matching multiplicities.* The same method rules out a G_2 explanation of -585 : its only admissible G_2 solution needs weight $2\omega_1$ with $T_2 = 99/4$ and the $\mathrm{SO}(3)$ -parameter Sym^2 of the CM form 9.4.a.a; the weight- $2\omega_1$ spectra at the three levels are $\{0, 81/2\}$, $\{0, 81/2\}$ and $\{0^3, 81/2^2, (-9 \pm \sqrt{1161})/4\}$, and in any case a form fixed by the parahoric Q_3 is Iwahori-spherical at 3, so every building block of its parameter has conductor dividing 3 and level-9 forms cannot occur.

8.2. The second prime on G_2 . The operator $T(\omega_1^\vee)(5)$ on G_2 (19530 neighbour maps) was computed on the same weight- $2\omega_2$ space at the line-parahoric level; it commutes with T_2 (to 10^{-10}) and the joint eigen-systems are

$$(T_2, T_5) = \left(\frac{9}{4}, \frac{14634}{25} \right), \quad \left(9, -\frac{11502}{125} \right)^{(2)}, \quad \left(-\frac{27}{2}, \frac{306}{5} \right)^{(2)},$$

with $T(\omega_1^\vee)(p) = p^3\chi_7(s_p) - 1$ in the unitary normalisation; the trivial-weight control at 5 returns $\{19530, 810\}$, the known values. The first eigen-system is a genuine second-prime test of the identification of the $9/4$ -form as the G_2 -lift of 3.8.a.a: the lift predicts $\chi_7(s_p) = (p^{1/2} + p^{-1/2})t_p + t_p^2 - 1$ with $t_p = a_p/p^{7/2}$, and $a_5(3.8.a.a) = 390$ gives $T_5 = 14634/25$, exactly the computed value. The two stable forms have $\chi_7(s_5) = -11377/15625$ and $311/625$, both in the tempered range.

Through $G_2 \times \mathrm{SO}(3)$ these give the predictions

$$T(\omega_1^\vee)(5) = 10439064 \text{ on the } 3087\text{-eigenspace,} \quad 13407624 \text{ on the } 567\text{-eigenspace,} \quad 23563224 \text{ on the } 2331\text{-eigenspace}$$

the last agreeing exactly with the prediction of the $\mathrm{Sp}_6 \times \mathrm{SL}_2$ route for 3.8.a.a[3] at 5. These are the values an operator of F_4 at 5 must return (Remark 1.1); its direct computation is out of reach, the number of $\omega_1^\vee$ -neighbours at 5 being $(5^{12} - 1)(5^4 + 1)/4 \approx 3.8 \cdot 10^{10}$ per class.

9. THE EIGENVALUE -585 AND THE GENUS OF E_7

9.1. Narrowing -585 . Before the computation of §9.2, three routes remained, and two were tested. (a) A genus-2 form F : over all pairings with newforms of level 1 and 3, the unique unitary solution with integral eigenvalues is F of weight $\mathrm{Sym}^6 \otimes \det^3$ with $\lambda_F(2) = -6$ and $\lambda_{\wedge^2}(2) = -352$ (or $\mathrm{Sym}^2 \otimes \det^6$, -12 , -1408), paired with 3.6.a.a. It is absent from the genus of A_5 at both of its levels, so its local component at 3 must be an Iwahori-spherical representation that does not transfer to the non-split $\mathrm{SO}(5)$; such a form is invisible to every compact inner form of GSp_4 split at 2, and deciding its existence requires holomorphic Siegel forms of level $\Gamma_0(3)$, which have not been tabulated (the computations of Bergström–Faber–van der Geer cover levels 1 and $\Gamma_0(2)$). (b) A stable Sp_6 -parameter: the shapes compatible with -585 with integral, tempered eigenvalues require $\mathrm{SO}(7)$ -weights $(1, 1, 0)$, $(2, 1, 0)$, $(2, 2, 0)$, $(3, 0, 0)$, $(2, 2, 2)$ or $(3, 1, 1)$. We computed compact $\mathrm{SO}(7)$ -forms on the genus of E_7 (one class, $|\mathrm{Aut}| = 2903040$; 63 two-neighbours with explicit isometries; 364 isotropic lines modulo 3 in two orbits; trivial weight gives $T_2 = 63$ and $\{63, 24\}$) in the first four weights, at level one and at the stabiliser of an isotropic line modulo 3: all spaces are 0 except one form of weight $(2, 2, 0)$ with $\lambda_1(2) = 24$, where -585 would need 36 or 39. (c) A stable form on F_4 . The weights $(2, 2, 2)$, $(3, 1, 1)$ and the deeper levels of $\mathrm{SO}(7)$ at 3 are computed in §9.2, which resolves -585 .

9.2. -585 identified. The two largest weights, $(2, 2, 2)$ (294-dimensional, in $\mathrm{Sym}^2 \wedge^3 V$) and $(3, 1, 1)$ (616-dimensional, in $\mathrm{Sym}^2 V \otimes \wedge^3 V$), and the deeper levels at 3—stabilisers of isotropic planes (3640, two orbits) and of isotropic 3-spaces (1120, one orbit) modulo 3—complete the $\mathrm{SO}(7)$ search. Weight $(3, 1, 1)$ has forms at both deeper levels, of dimensions 2 and 1, with T_2 giving $\lambda_1(2) = -60$: the first entry of the list of integral, tempered solutions for -585 in that shape, $(\lambda_1, \lambda_2, \lambda_3) = (-60, 3264, -79872)$. We then built the other two Hecke operators at 2 on the genus of E_7 : the operator of the 630 neighbours along totally isotropic planes modulo 2, calibrated by the constant function as $T_P = 16\chi_{\wedge_0^2} - 5$, and the operator of the 1080 neighbours along the 135 maximal totally isotropic 3-spaces, $T_3 = 2^{9/2}\chi_{\wedge_0^3} - 2^{5/2}\chi_{\mathrm{std}}$; both commute with T_2 . On the $(3, 1, 1)$ -forms they give $T_P = 9/2$, i.e. $\mathrm{tr} \wedge_0^2 F(s_2) = 19/32$, i.e. $\lambda_2 = 3264$; and $T_3 = 18$, i.e. $\lambda_3 = -79872$. All three invariants of the Satake parameter at 2 agree with the prediction, so

$$-585 : \quad \psi = \wedge_0^2 F \oplus F[2],$$

with F the Sp_6 -parameter of a compact $\mathrm{SO}(7)$ -form of weight $(3, 1, 1)$ and level 3, with Hodge weights $\pm \frac{11}{2}, \pm \frac{5}{2}, \pm \frac{3}{2}$, reproduces both F_4 -eigenvalues $(-585, 9000)$. Its Frobenius polynomial at 2 shows that F is endoscopic, $F = 3.6.a.a \oplus G$ with G a stable genus-2 form (§10). With this every one of the fifteen dimensions at level Q_3 is identified in the sense of Remark 1.1, each by two or more independent Hecke operators on a smaller group.

10. STABILITY

From the three (resp. two) Hecke eigenvalues at 2 one recovers the full Frobenius polynomial at 2 of each new parameter, in the arithmetic normalisation, and its factorisation over $\mathbb{Q}$ decides endoscopy at the level of the Satake parameter: an endoscopic parameter's polynomial factors

over the Hecke fields of its pieces, while a Galois group equal to the full Weyl group forbids every factorisation that the relevant endoscopic shapes would impose. As a control, the weight- $2\omega_2$ G_2 -form with $T_2 = 9/4$ (the lift of 3.8.a.a) gives $(X^2 - 96X + 2^{15})(X^2 - 48X + 2^{13})(X^2 + 220X + 2^{14})$, with roots off the unitary circle, as a non-tempered lift must.

- *The G_2 -forms behind 3087 and 567.* Their Frobenius sextics $X^6 - 32X^5 + 18176X^4 - 720896X^3 + \dots + 2^{42}$ and $X^6 + 328X^5 + 58496X^4 + 7757824X^3 + \dots + 2^{42}$ are irreducible, with Galois group $D_6 = W(G_2)$, and all roots of absolute value 2^7 . The Arthur shapes are excluded by temperedness; the SL_3 -shape $1 \oplus \pi \oplus \pi^\vee$ would need π of Hodge type $\{7, -3, -4\}$, excluded by Clozel's purity lemma; the $\mathrm{SO}(4)$ -shapes need newforms of weight 4 and 12, and there are none of weight 4 and level dividing 3. *No endoscopic parameter of these shapes reproduces them; we call the forms stable in this sense.*
- *The $\mathrm{SO}(7)$ -form behind -585 is not stable.* Its Frobenius polynomial $X^6 + 60X^5 + 3264X^4 + 79872X^3 + 2^{11} \cdot 3264X^2 + 2^{22} \cdot 60X + 2^{33}$ factors as $(X^2 + 48X + 2^{11})(X^4 + 12X^3 + 640X^2 + 24576X + 2^{22})$. The quadratic is 3.6.a.a ($8a_2 = -48$), and the quartic is a genus-2 parameter G with $\lambda_G(2) = -12$ and $\mathrm{tr} \wedge_0^2 G(s_2) = 640/2^{11} - 1 = -11/16$: exactly the genus-2 form predicted for -585 in §4.1, in its weight- $\mathrm{Sym}^2 \otimes \det^6$ version, paired with exactly the predicted newform. So the parameter of -585 is $\wedge_0^2 G \oplus G \otimes 3.6.a.a \oplus [1] \oplus G[2] \oplus 3.6.a.a[2]$, the genus-2 shape of §4.1. The form G is absent from the genus of A_5 because its local component at 3 does not transfer to the non-split $\mathrm{SO}(5)$; it becomes visible as the Sp_4 -part of an endoscopic form on the split $\mathrm{SO}(7)$ of E_7 .
- *The genus-2 parameter G is stable in the same sense.* Its quartic is irreducible, with Galois group D_4 , and tempered; a Yoshida lift would need newforms of weights 12 and 4, and a Saito–Kurokawa lift is non-tempered.

Thus the level- Q_3 module of F_4 contains, besides lifts of classical newforms, three stable objects on smaller groups that had not been computed before: two compact G_2 -forms of weight $2\omega_2$ and level 3, and a stable parameter G of GSp_4 of weight $\mathrm{Sym}^2 \otimes \det^6$ and level 3 whose Hecke eigenvalue at 2 was predicted from F_4 before it was found. We have established G only as a factor of the Frobenius polynomial of an $\mathrm{SO}(7)$ -form; Arthur's classification for the split $\mathrm{SO}(7)$, in Taïbi's extension to inner forms compact at infinity, should realise it as a holomorphic Siegel cusp form of level 3, and we have not carried out that step.

Question 10.1. What are the Galois representations of the stable forms of §10, and which L -values do their congruences involve? What explains the remaining cuspidal congruences $567 \equiv \{15057, 3087\} \pmod{5}$, $567 \equiv 2331 \pmod{7}$ and $207 \equiv -585 \pmod{11}$ in terms of these parameters?

11. CONCLUDING REMARKS

Every one of the fifteen dimensions of the level- Q_3 module of F_4 is now matched with a parameter on a smaller group, in the sense of Remark 1.1, and every congruence in the module is accounted for by the organising principle of §5. Three of the parameters are new: two compact G_2 -forms of weight $2\omega_2$ and level 3, and a genus-2 parameter of weight $\mathrm{Sym}^2 \otimes \det^6$ and level 3 that is invisible on the non-split $\mathrm{SO}(5)$ and appears as the Sp_4 -part of an endoscopic form on the split $\mathrm{SO}(7)$. What remains is a confirmation at a second prime, which requires a Hecke operator of F_4 at 5 (the neighbour count there is of the order of $4 \cdot 10^{10}$ per class, so a different method than direct enumeration is needed), the realisation of the genus-2 parameter as a holomorphic Siegel form through Arthur's classification, and the Galois representations of the three new forms.

APPENDIX A. THE TWO HECKE MATRICES AT LEVEL Q_3

Classes 1–9 are on $J_{\mathbb{Z}}$ and 10–15 on J_E , in the order of the stabiliser lists above. Entry (i, j) counts the $\omega_1^\vee$ -neighbours at 2 of class i lying in class j .

18270	3528	2268	1344	189	189	42	0	0	92232	10920	5040	3528	1344	336
15876	7011	0	2808	0	0	135	0	0	93744	9504	5400	3024	1296	432
15309	0	9279	0	567	675	0	0	0	91692	11340	3888	4212	2268	0
14112	6552	0	4851	0	0	252	63	0	84672	19656	7560	0	756	756
10206	0	4536	0	10521	378	0	0	189	81648	18144	0	9072	4536	0
15309	0	8100	0	567	1854	0	0	0	83592	13608	7776	8424	0	0
10584	7560	0	6048	0	0	1449	189	0	84672	12096	3024	0	4536	9072
0	0	0	13440	0	0	1680	8064	2646	0	94080	0	0	5880	13440
0	0	0	0	15120	0	0	5103	5607	0	90720	0	0	22680	0
11529	2604	1698	1008	189	129	42	0	0	99945	11214	5568	3582	1386	336
10920	2112	1680	1872	336	168	48	42	21	89712	21015	5280	3024	2376	624
11340	2700	1296	1620	0	216	27	0	0	100224	11880	8415	864	108	540
11907	2268	2106	0	567	351	0	0	0	96714	10206	1296	10413	3402	0
10752	2304	2688	576	672	0	144	21	42	88704	19008	384	8064	5679	192
10584	3024	0	2268	0	0	1134	189	0	84672	19656	7560	0	756	9387

The matrix of $T(\omega_4^\vee)(2)$, in the same ordering (row sums 8,910,720):

851355	186480	122796	76800	13473	10233	2940	162	81	6196392	758136	343296	230760	94296	23520
839160	222876	81648	101304	6804	6804	5481	243	0	6223824	743040	359208	207792	84888	27648
828873	122472	220608	46656	26811	17199	1458	0	243	6164748	761724	311040	276372	120852	11664
806400	236376	72576	129348	6048	6048	5004	2142	378	6035904	926856	406728	145152	86940	44820
727542	81648	214488	31104	183618	17874	972	2187	4887	5820336	1013472	163296	428976	212544	7776
828873	122472	206388	46656	26811	31419	1458	0	243	6218856	795096	252720	280584	87480	11664
740880	306936	54432	120096	4536	4536	25830	5940	1134	5927040	1064448	311472	108864	72576	162000
362880	120960	0	456960	90720	0	52800	126324	53676	2903040	3743040	241920	0	336000	422400
349920	0	155520	155520	390960	12960	19440	103518	76482	2799360	3667680	0	311040	712800	155520
774549	172884	114162	71856	13473	9597	2940	162	81	6289722	762462	349584	230814	94914	23520
758136	165120	112848	88272	18768	9816	4224	1671	849	6099696	940392	342816	220752	114480	32880
772416	179604	103680	87156	6804	7020	2781	243	0	6292512	771336	401184	184896	70740	30348
778815	155844	138186	46656	26811	11691	1458	0	243	6231978	745038	277344	347454	137538	11664
754368	150912	143232	66240	31488	8640	2304	1200	1320	6074496	915840	251520	326016	161064	22080
740880	193536	54432	134460	4536	4536	20250	5940	1134	5927040	1035720	424872	108864	86940	167580

APPENDIX B. DATA AND CODE

All code (Python/NumPy, FLINT for exact characteristic polynomials), the class sets and Hecke matrices of F_4 at levels 1 and Q_3 , the G_2 computations in weights $2\omega_1$ and $2\omega_2$, and the $SO(5)$ and $SO(7)$ computations on the genera of A_5 and E_7 are available from the author.

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